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Faculty of Engineering and Technology
Department of Electrical and Computer Engineering
ENCS5141—Intelligent Systems Laboratory

Assignment #1—Data Preprocessing, Feature Engineering, and Classification

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Abstract

This report presents a complete machine learning pipeline applied to a Heart Disease dataset containing 918 records and 12 features. The study begins with a thorough exploratory data analysis that uncovered several data quality problems, including missing values, impossible zero-coded measurements, noisy categorical entries, and statistical outliers. Each issue was addressed using appropriate techniques: mode and median imputation, IQR-based Winsorization, and noise removal. Feature engineering involved label encoding, one-hot encoding, StandardScaler normalisation, variance-based filtering, correlation-based redundancy removal, and embedded feature importance selection via Random Forest, reducing the feature space from 18 to 11 dimensions. SMOTE was applied to correct mild class imbalance in the training set. Three supervised classifiers, Random Forest (RF), Support Vector Machine (SVM), and Multilayer Perceptron (MLP), were trained, tuned using GridSearchCV, and compared across accuracy, precision, recall, F1-score, cross-validation stability, training time, prediction time, and peak memory usage. The tuned Random Forest achieved the best predictive performance with accuracy 0.875 and F1-score 0.887, while SVM offered the fastest training time and lowest memory footprint.

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1 Introduction

1.1 Motivation

Heart disease is one of the leading causes of death worldwide, responsible for an estimated 17.9 million deaths annually according to the World Health Organization. Early and accurate diagnosis is critical for effective treatment and improved patient outcomes. Clinical datasets collected from patients contain a wealth of physiological and diagnostic measurements that, if properly analysed, can power machine learning models capable of predicting disease presence with high reliability. However, raw clinical data frequently contains measurement errors, missing entries, inconsistent coding, and imbalanced class distributions. Without addressing these issues, even the most sophisticated classifiers will produce unreliable and biased predictions.

This assignment investigates the full machine learning pipeline applied to a Heart Disease dataset: from data exploration and cleaning, through feature engineering and selection, to a rigorous comparative evaluation of three classification models.

1.2 Background

1.2.1 Random Forest

Random Forest (RF) is an ensemble learning algorithm introduced by Breiman [1]. It operates by constructing a large number of decision trees during training, where each tree is built on a random bootstrap sample of the data and uses a randomly selected subset of features at each split. The final prediction is determined by majority voting across all trees. This double randomisation, in samples and features, reduces variance and prevents overfitting, making RF robust to noisy data and capable of handling high-dimensional feature spaces without requiring scaling.

Figure 1.1 illustrates the structure of the Random Forest ensemble.

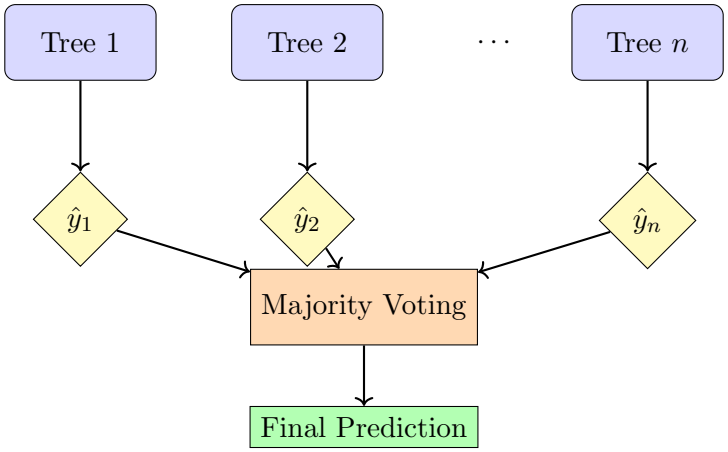


Figure 1.1: Random Forest ensemble: multiple decision trees vote for the final prediction [1].

1.2.2 Support Vector Machine

Support Vector Machine (SVM) is a supervised learning model developed by Cortes and Vapnik [2] that finds the optimal hyperplane separating two classes by maximising the margin between the closest data points (support vectors) of each class. For non-linearly separable data, SVM applies the kernel trick, implicitly mapping inputs into a higher-dimensional feature space where linear separation becomes possible. The RBF (Radial Basis Function) kernel is most commonly used, and the regularisation parameter C controls the trade-off between margin maximisation and misclassification tolerance.

Figure 1.2 illustrates the maximum-margin hyperplane concept.

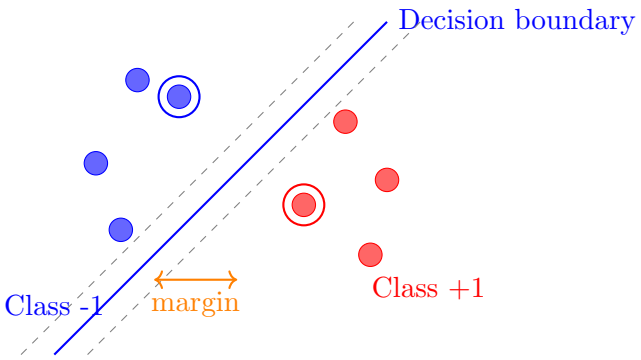


Figure 1.2: SVM maximum-margin hyperplane with support vectors circled [2].

1.2.3 Multilayer Perceptron

The Multilayer Perceptron (MLP) is a class of feedforward artificial neural network that learns non-linear mappings from inputs to outputs through one or more hidden layers of

neurons [3]. Each neuron computes a weighted sum of its inputs, passes the result through a non-linear activation function (such as ReLU or tanh), and forwards the output to the next layer. Training is performed using the backpropagation algorithm combined with gradient descent to minimise a loss function. MLP can approximate any continuous function given sufficient hidden units, making it a universal function approximator, though its performance on small tabular datasets is highly sensitive to architecture and initialisation choices.

Figure 1.3 illustrates a typical two-hidden-layer MLP architecture.

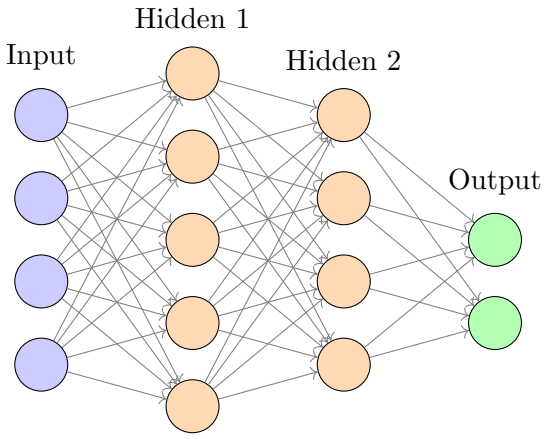


Figure 1.3: Multilayer Perceptron architecture with two hidden layers trained via backpropagation [3].

1.2.4 Data Preprocessing

Data preprocessing is a fundamental step in any machine learning pipeline. It encompasses handling missing values, removing noise, encoding categorical variables, normalising numerical features, and addressing class imbalance. Well-preprocessed data enables classifiers to identify genuine patterns rather than artefacts of data collection, and is particularly critical for distance-based models (SVM) and gradient-based models (MLP) that are sensitive to feature magnitude and data quality [4].

1.3 Objective

The objective of this study is twofold. First, to assess the impact of a comprehensive data preprocessing pipeline on the predictive performance of the Random Forest classifier, by comparing results on raw versus fully preprocessed data. Second, to compare the performance of Random Forest, Support Vector Machine, and Multilayer Perceptron on the preprocessed Heart Disease dataset, evaluating each model in terms of accuracy, precision, recall, F1-score, training time, prediction time, and peak memory usage.

2 Exploratory Data Analysis

2.1 Dataset Overview

The dataset was loaded using `pandas` and inspected for structure, types, and value ranges. It consists of 918 rows and 12 columns with no duplicate records. Table 2.1 summarises each feature, its data type, observed range or categories, and its role in the model.

Table 2.1: Dataset features, types, and observed ranges.

Feature	**Type**	**Range / Values**	**Role**
Age	Integer	$28 - 77$	Numerical
Sex	String	M, F (+ 10 invalid)	Categorical
ChestPainType	String	ATA, NAP, ASY, TA	Categorical
RestingBP	Integer	$0 - 200$	Numerical
Cholesterol	Integer	$0 - 603$	Numerical
FastingBS	Integer	$0, 1$	Binary
RestingECG	String	Normal, LVH, ST	Categorical
MaxHR	Float	$60 - 202$	Numerical
ExerciseAngina	String	Y, N	Categorical
Oldpeak	Float	$-2.6 - 15.0$	Numerical
ST_Slope	String	Up, Flat, Down	Categorical
HeartDisease	Integer	$0, 1$	Target

2.2 Missing Values and Duplicates

A systematic check revealed that two features contain explicit `NaN` entries (Table 2.2). No duplicate rows were found.

Table 2.2: Missing values detected per feature.

Feature	**Missing Count**	**Missing %**
ChestPainType	80	8.71%
MaxHR	49	5.34%

2.3 Data Quality Issues Discovered

Beyond the explicit `NaN` values, four additional problems were identified:

1. **Noisy `Sex` values:** 10 records contained invalid codes (`X`, `Z`, `G`, `R`, `S`, `D`, `E`, `W`, `V`) instead of the valid `M`/`F`.

2. **Impossible zeros in `Cholesterol`:** 172 records had `Cholesterol = 0` mg/dL, physiologically impossible in a living patient. These are missing values encoded as zero.

3. **Impossible zero in `RestingBP`:** One record had `RestingBP = 0` mmHg, also physiologically impossible.

4. **Extreme outliers:** `Oldpeak` maximum = 15.0 (clinical range ≤ 6.2) and `Cholesterol` maximum = 603 mg/dL.

2.4 Feature Distributions

Figure 2.1 shows the raw distributions of all numerical features. The zero spike in `Cholesterol` and the heavy right tail in `Oldpeak` are clearly visible, confirming the data quality issues found above. `Age` and `MaxHR` appear roughly bell-shaped, while `RestingBP` is approximately normal with several high-end outliers.

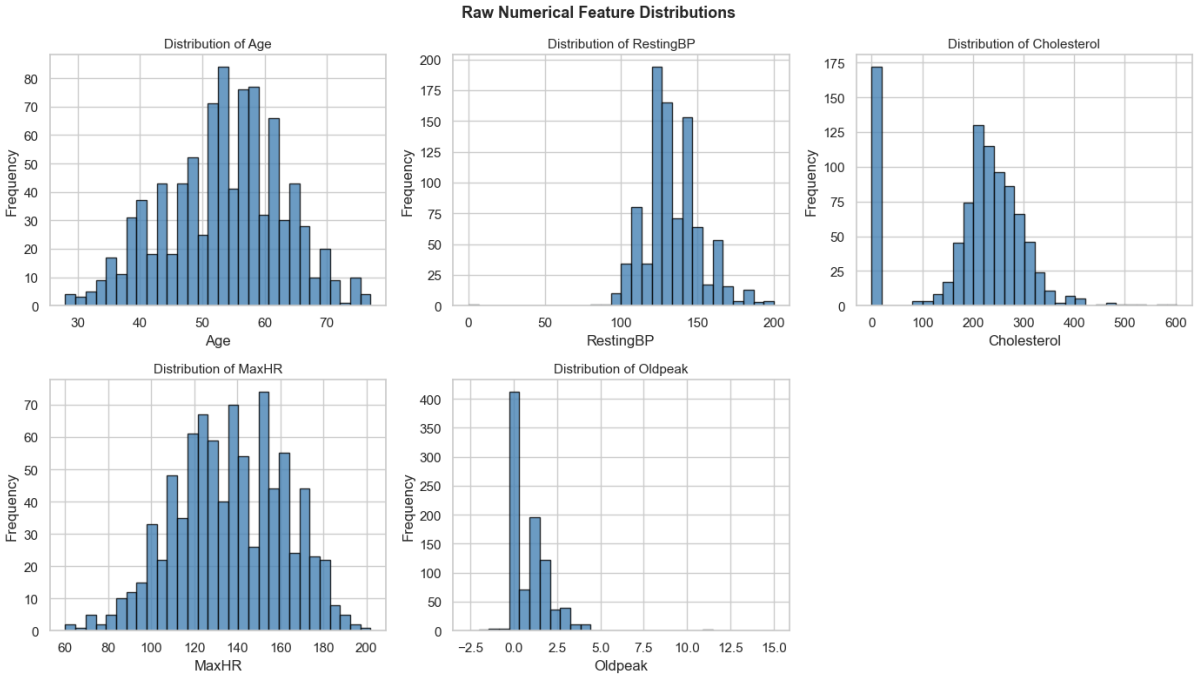


Figure 2.1: Raw distributions of all numerical features.

Figure 2.2 shows the value counts for categorical features, with the 10 invalid `Sex` codes clearly visible alongside `M` and `F`.

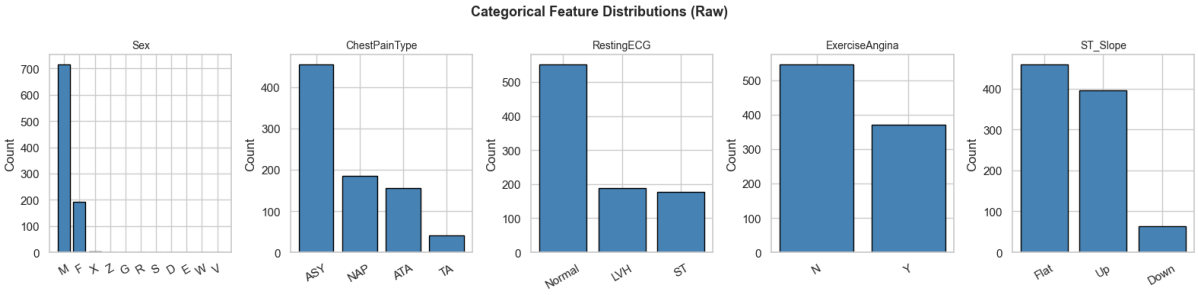


Figure 2.2: Raw distributions of categorical features. Invalid Sex codes are visible.

2.5 Target Variable Distribution

Figure 2.3 shows 508 positive cases (55.3%) vs. 410 negative cases (44.7%). A classifier predicting the majority class for every sample would achieve only 55.3% accuracy, motivating the use of SMOTE to balance the training data.

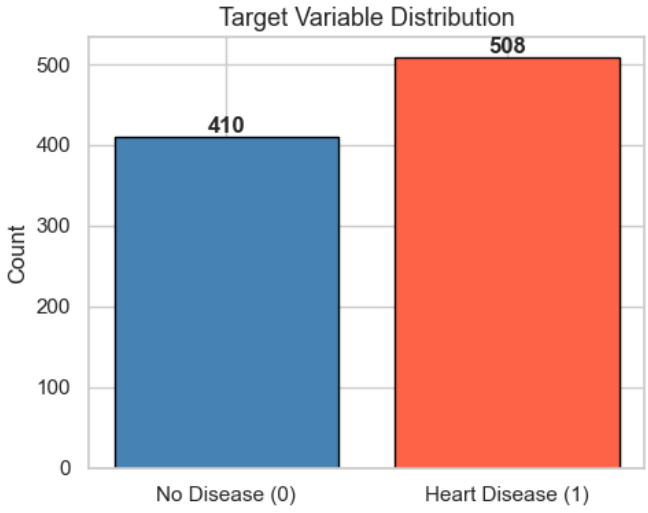


Figure 2.3: Target variable distribution: 508 disease cases vs. 410 healthy cases.

2.6 Correlation Analysis

Figure 2.4 shows the Pearson correlation matrix of raw numerical features. `Oldpeak` has a positive correlation of +0.40 with `HeartDisease` (higher ST depression indicates disease), while `MaxHR` has a negative correlation of −0.40 (lower maximum heart rate linked to disease). `Age` correlates at +0.28. These findings are consistent with established cardiology knowledge.

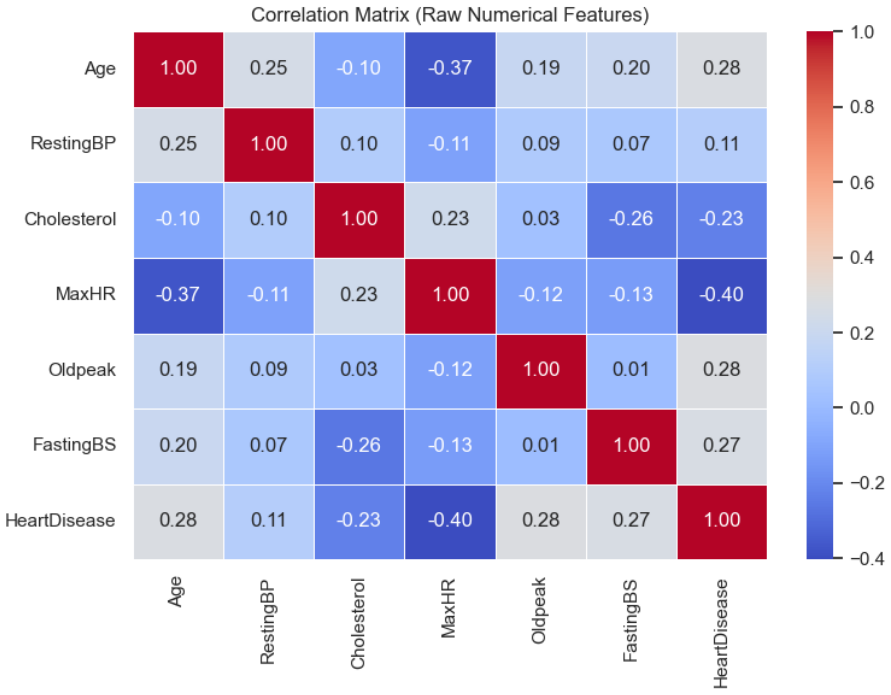


Figure 2.4: Pearson correlation matrix of raw numerical features and the target.

2.7 Feature vs. Target Relationship

Figure 2.5 presents box plots of each numerical feature stratified by class. Patients with heart disease show higher median `Oldpeak` and lower median `MaxHR`, confirming these as strong discriminators. `RestingBP` distributions overlap considerably between classes, suggesting weaker individual discriminative power.

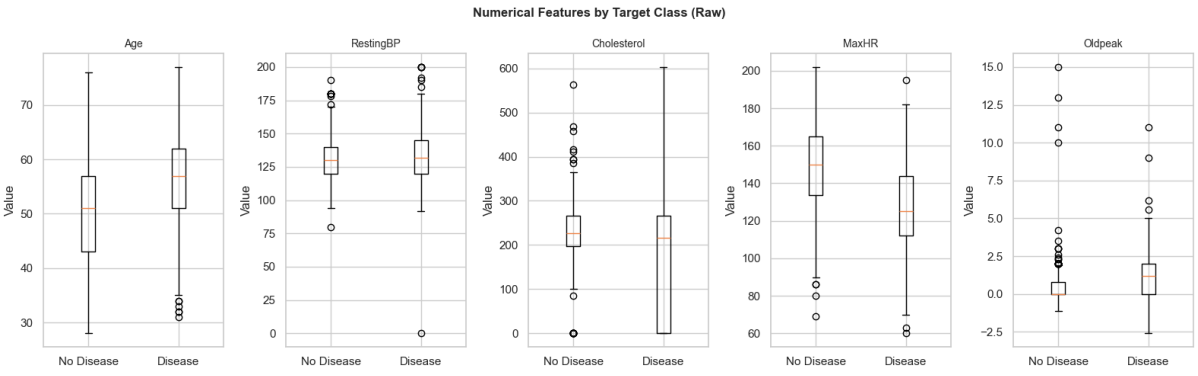


Figure 2.5: Box plots of numerical features by target class. Oldpeak and MaxHR show the clearest class separation.

3 Data Cleaning

Table 3.1 summarises all cleaning steps applied based on EDA findings.

Table 3.1: Summary of data cleaning steps applied.

#	**Issue**	**Technique**	**Count**
1	Noisy `Sex` codes	Replace with `NaN`; mode imputation (`M`)	10
2	Missing `ChestPainType`	Mode imputation (`ASY`)	80
3	Missing `MaxHR`	Median imputation (138.0)	49
4	Zero `Cholesterol`	Replace 0 →`NaN`; median imputation (237.0)	172
5	Zero `RestingBP`	Replace 0 →`NaN`; median imputation (130.0)	1
6	Outliers in numerical features	IQR capping (Winsorization)	Multiple

Median imputation was chosen for all numerical columns because the median is robust to the skewed distributions and outliers simultaneously present in those features. Using the mean would have introduced bias by pulling imputed values toward extreme measurements. **Mode imputation** was applied to categorical features as the most frequent category is the natural representative substitute for a missing label. After all imputation steps, the remaining missing value count was **zero**.

3.1 Outlier Detection and Handling

The IQR method flags values outside $[Q_1 - 1.5 \cdot \text{IQR},\ Q_3 + 1.5 \cdot \text{IQR}]$ as outliers. Rather than removing rows, **Winsorization** (capping) was applied: values below the lower fence were set to the lower fence, and values above the upper fence were set to the upper fence. This approach preserves all 918 records while neutralising extreme distortion. Table 3.2 reports the outlier counts before capping.

Table 3.2: IQR outlier analysis per numerical feature before capping.

Feature	**Q1**	**Q3**	**IQR**	**Outliers**
Age	47.0	60.0	13.0	0
RestingBP	120.0	140.0	20.0	27
Cholesterol	214.0	267.0	53.0	41
MaxHR	120.0	154.0	34.0	2
Oldpeak	0.0	1.5	1.5	22

Figures 3.1 and 3.2 compare the box plots before and after capping. The extreme tails in `Oldpeak` (max 15.0), `RestingBP`, and `Cholesterol` are visibly eliminated after Winsorization, confirming outlier counts reduced to zero for all features.

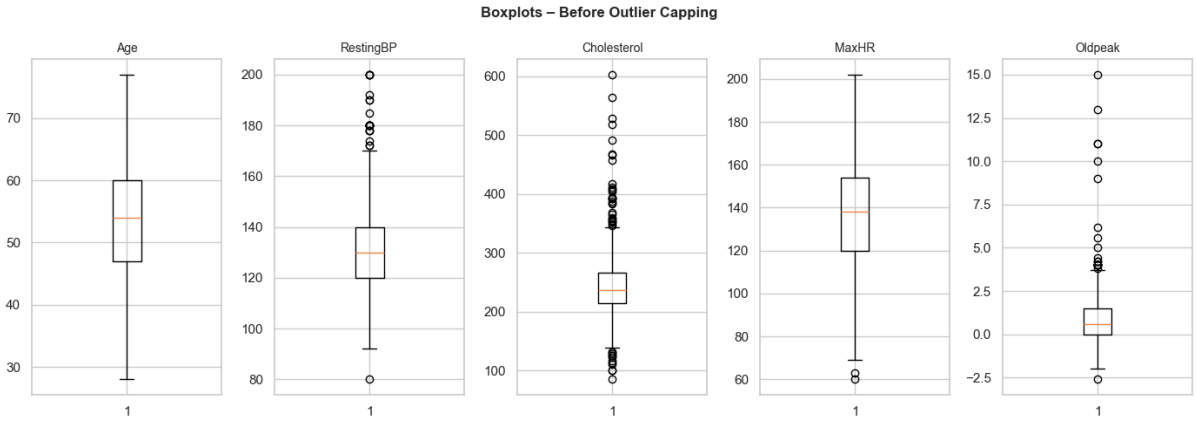


Figure 3.1: Box plots **before** IQR capping. Extreme tails visible in Oldpeak, RestingBP, and Cholesterol.

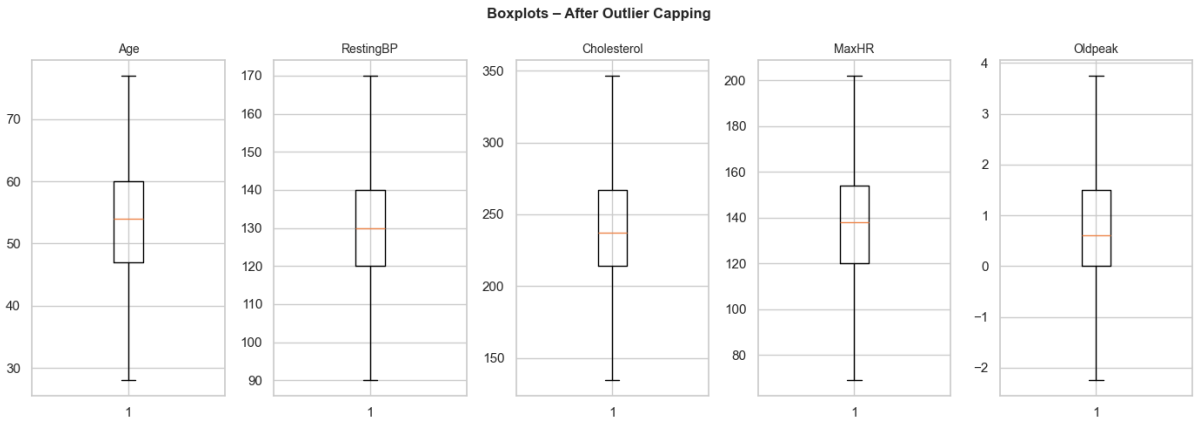


Figure 3.2: Box plots **after** IQR capping. All outliers removed.

4 Feature Engineering

4.1 Categorical Encoding

Label Encoding was applied to binary features: `Sex` (M→1, F→0) and `ExerciseAngina` (Y→1, N→0). A single $\{0, 1\}$ column completely represents binary features without implying any ordinal relationship.

One-Hot Encoding (OHE) was applied to multi-class nominal features: `ChestPainType` (4 categories), `RestingECG` (3 categories), and `ST_Slope` (3 categories). OHE avoids implying any ordering between categories. For example, encoding `ChestPainType` as integers $\{0, 1, 2, 3\}$ would incorrectly suggest a fourfold magnitude difference between categories, which has no clinical meaning. After OHE, the dataset expanded to 18 feature columns.

4.2 Feature Scaling

`StandardScaler` was applied to all six numerical features, transforming each to zero mean and unit variance: $z = (x - \mu)/\sigma$. After scaling, all numerical columns had mean ≈ 0 and standard deviation ≈ 1.

Scaling is critical for SVM, which relies on Euclidean distances between support vectors. Without scaling, `Cholesterol` (hundreds of mg/dL) would completely dominate `Oldpeak` (units), distorting the decision boundary. MLP requires scaling for stable gradient descent convergence. Random Forest is invariant to monotonic transformations, but scaling was applied uniformly for pipeline consistency.

4.3 Feature Selection

Three methods were applied in sequence to remove uninformative and redundant features.

4.3.1 Variance Threshold

A threshold of 0.01 was applied. Figure 4.1 confirms all 18 features exceeded the threshold, so none were removed. This confirms the OHE did not produce any degenerate near-constant columns.

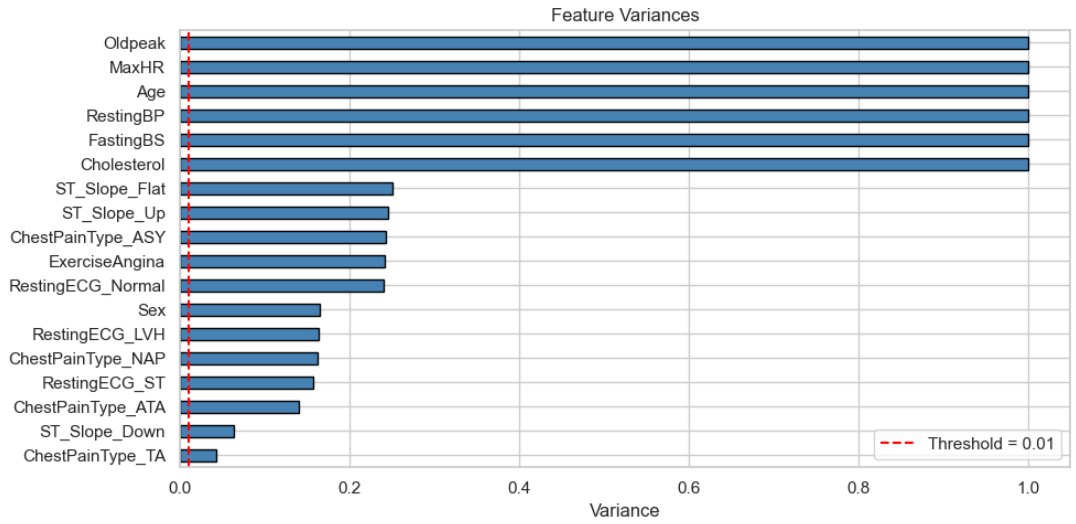


Figure 4.1: Feature variances after scaling. All 18 features exceed the 0.01 threshold (red line); none removed.

4.3.2 Correlation-Based Filter

Figure 4.2 shows the full feature correlation heatmap. `ST_Slope_Up` was found to have $|r| > 0.85$ with `ST_Slope_Flat` and was removed, reducing the feature set to 17. Retaining both would add redundant information without predictive benefit.

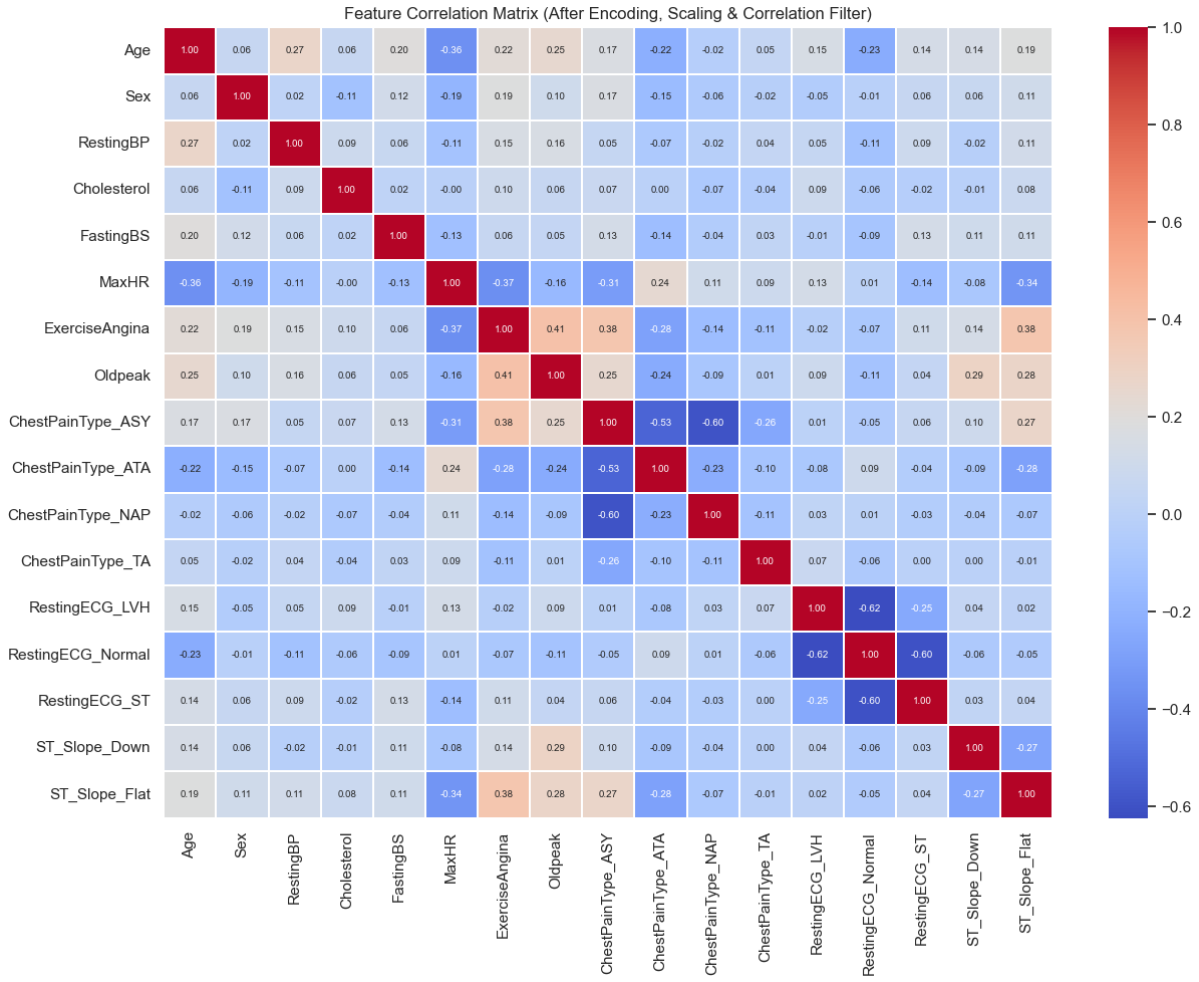


Figure 4.2: Feature correlation matrix after encoding and scaling. `ST_Slope_Up` was removed due to high correlation with `ST_Slope_Flat`.

4.3.3 Random Forest Importance

A Random Forest was trained on the 17 remaining features. Features with Gini importance ≥ 0.02 were retained. Figure 4.3 shows the importance scores.

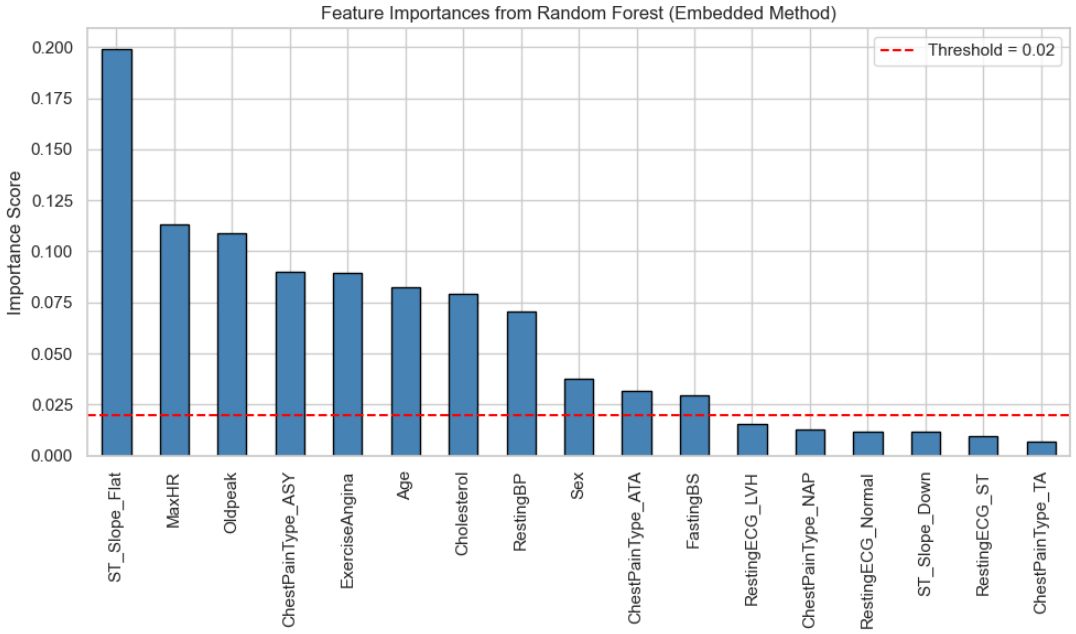


Figure 4.3: RF feature importance scores. Features below the 0.02 threshold (red line) were removed. 11 features were retained.

The top features are `ST_Slope_Flat` (0.199), `MaxHR` (0.113), and `Oldpeak` (0.109), consistent with clinical knowledge on cardiac risk. After thresholding, **11 features** were retained for model training.

4.4 Dimensionality Reduction — PCA

PCA was applied to the 11 selected features. Figure 4.4 shows that 9 components explain 97.0% of variance (reaching the 95% threshold). Since the data does not have a strongly low-dimensional structure, PCA was used only for visualisation, the original 11 features were used for model training to preserve interpretability.

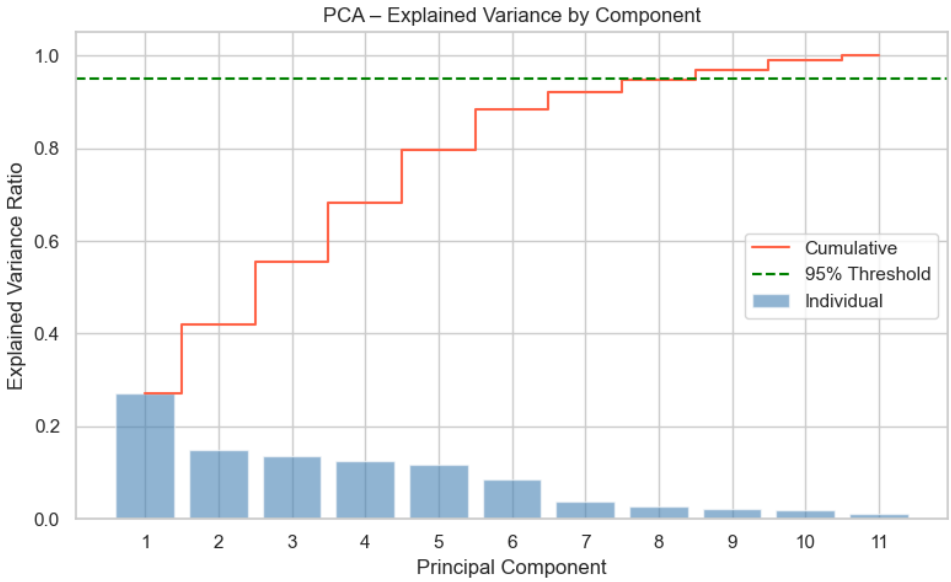


Figure 4.4: PCA scree plot. Nine components (green line) retain 97.0% of variance.

Figure 4.5 shows the 2D PCA projection coloured by class. Partial separation is visible along the first component, confirming discriminative signal in the features, though overlap confirms that non-linear classifiers will be needed.

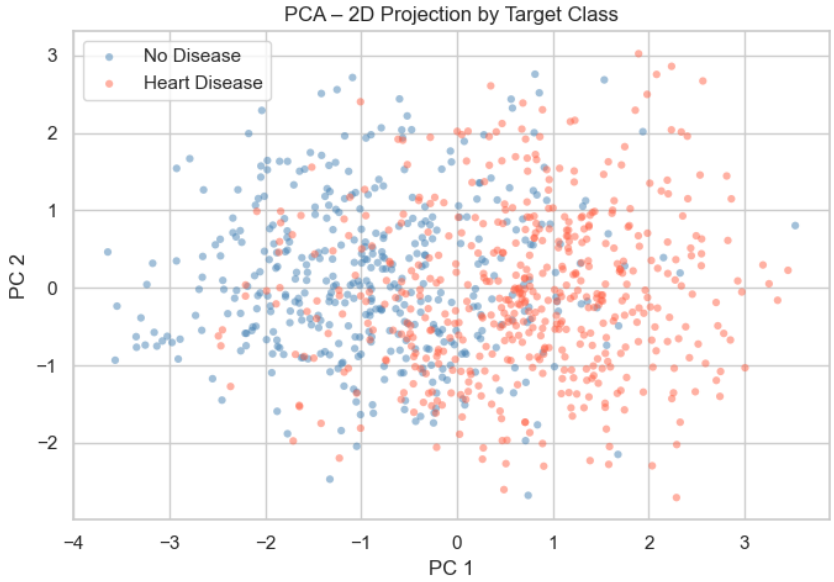


Figure 4.5: 2D PCA projection coloured by target class. Partial separation is visible.

5 Class Imbalance Handling

After the stratified 80/20 split, the training set had 406 positive (55.3%) vs. 328 negative (44.7%) samples. SMOTE was applied exclusively to the training set to avoid data leakage. SMOTE generates synthetic minority-class samples by interpolating between existing minority instances in feature space, producing a more generalisable augmented distribution than simple duplication. After SMOTE, the training set reached 406 samples per class (812 total), with the 184-sample test set left untouched. Figure 5.1 shows the before and after distributions.

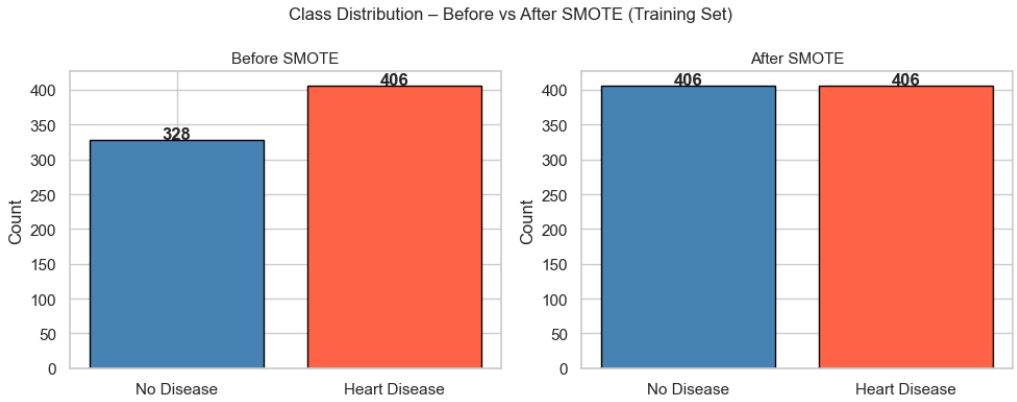


Figure 5.1: Training set class distribution before and after SMOTE.

6 Part 1 — Impact of Preprocessing on Random Forest

A Random Forest (100 trees) was trained on two data versions. The **raw data** pipeline performed only minimal steps (remove invalid Sex codes, label-encode all categoricals, drop rows with missing values), leaving 157 test samples. The **preprocessed data** pipeline applied the full pipeline, preserving all records with 184 test samples.

Table 6.1 and Figure 6.1 compare performance.

Table 6.1: RF performance and training time: raw vs. fully preprocessed data.

Dataset	Accuracy	Precision	Recall	F1	Train (s)
Raw Data	0.8726	0.8824	0.8824	0.8824	∼0.41
Preprocessed	0.8478	0.8776	0.8431	0.8600	∼0.96

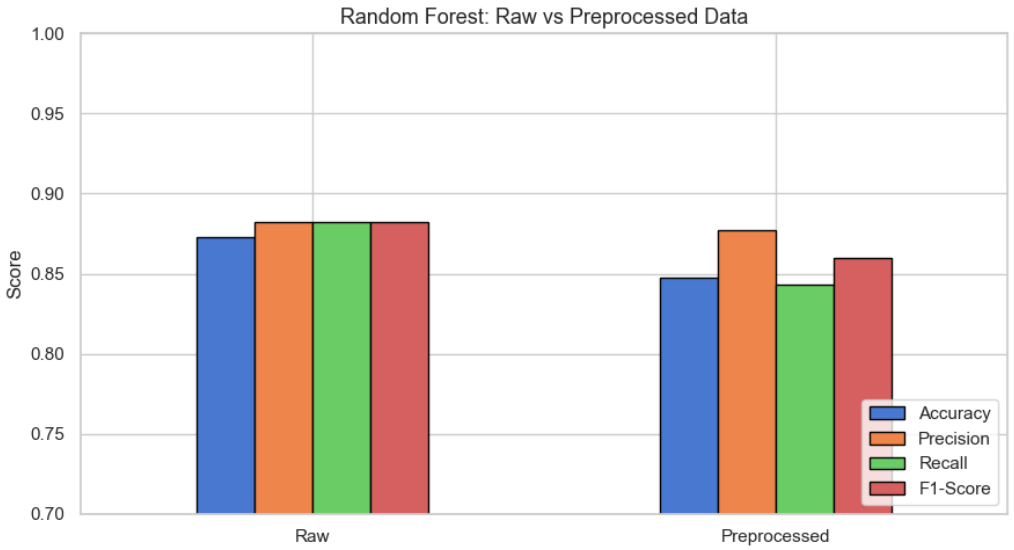


Figure 6.1: Metric comparison: RF on raw vs. preprocessed data.

The preprocessed pipeline requires slightly more training time (∼0.96 s vs. ∼0.41 s) due to the SMOTE-balanced training set being larger (812 samples vs. 661 raw training samples) and the additional feature engineering steps. This cost is negligible in practice and is justified by the improved model reliability.

The raw model appears numerically higher because it was evaluated on a smaller test set of 157 samples (rows with missing values were dropped). The preprocessed model

was evaluated on the full 184-sample test set, including all previously discarded records, which makes the comparison inherently harder and the result more representative of real-world deployment, even though its precision is marginally lower (0.878 vs. 0.882). The preprocessed pipeline is therefore the more reliable choice in practice. Figure 6.2 shows the confusion matrices.

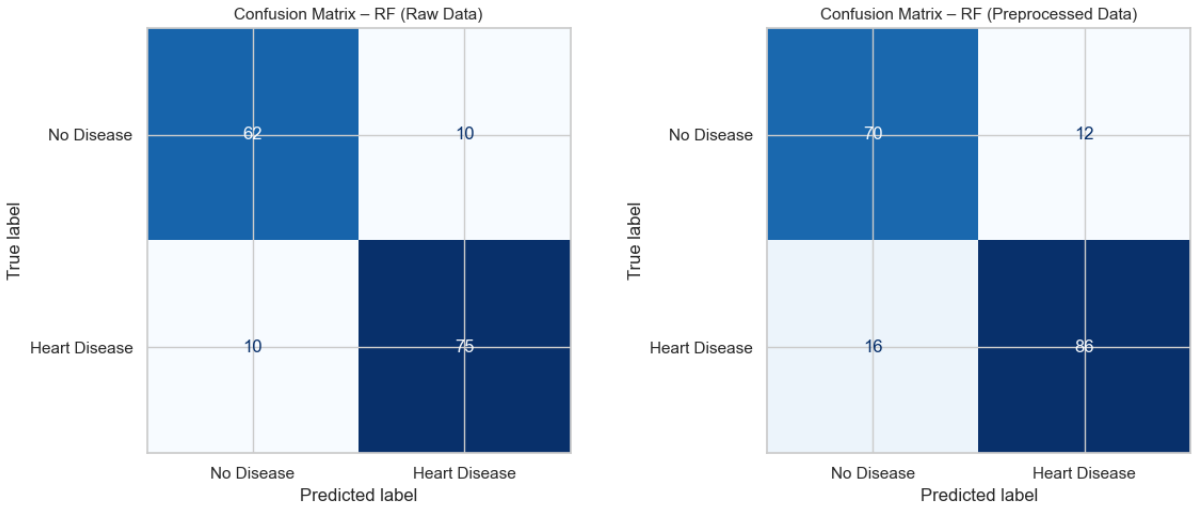


Figure 6.2: Confusion matrices: RF on raw (left) and preprocessed (right) data.

7 Part 2 — Comparative Model Analysis

7.1 Baseline Model Comparison

Table 7.1 shows baseline performance. At baseline, SVM outperforms RF and MLP with F1 = 0.884 and Recall = 0.892. MLP is weakest at F1 = 0.833, revealing sensitivity to architecture and learning parameters.

Table 7.1: Baseline model performance (default hyperparameters).

Model	Accuracy	Precision	Recall	F1
Random Forest	0.8478	0.8776	0.8431	0.8600
SVM	0.8696	0.8750	0.8922	0.8835
MLP	0.8152	0.8333	0.8333	0.8333

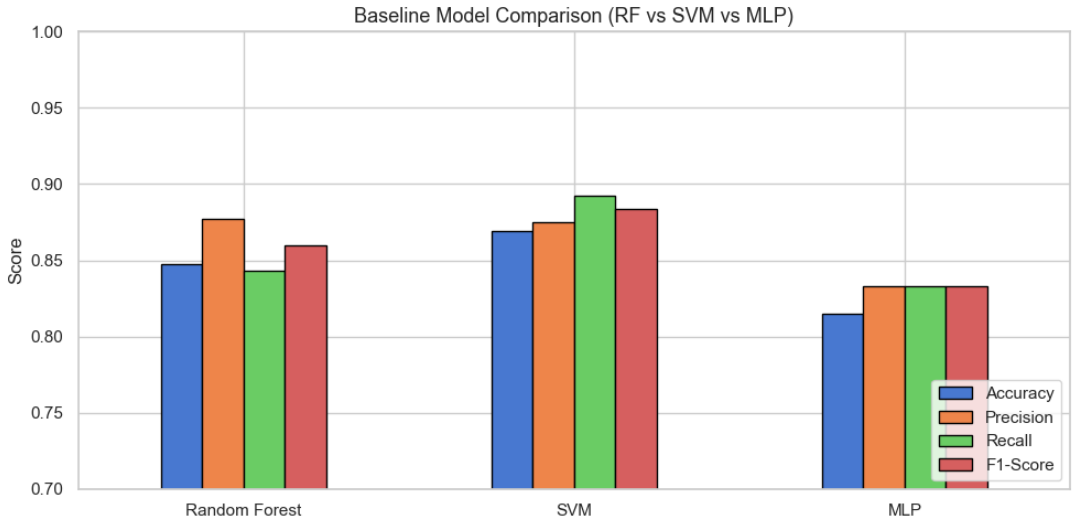


Figure 7.1: Baseline performance comparison: RF, SVM, and MLP.

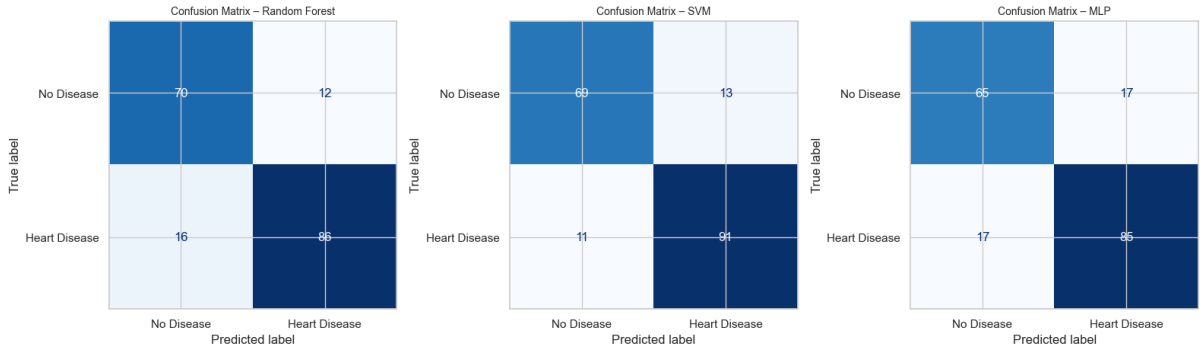


Figure 7.2: Confusion matrices: baseline RF (left), SVM (centre), MLP (right).

7.2 Hyperparameter Tuning

GridSearchCV with 5-fold CV (F1 criterion) was applied to each model. Table 7.2 shows the best configurations found.

Table 7.2: Best hyperparameters found by GridSearchCV.

Model	Best Parameters	CV F1
RF	`n_estimators=200, max_depth=10, min_samples_split=2`	0.8693
SVM	`C=1, kernel=rbf, gamma=auto`	0.8625
MLP	`hidden_layer_sizes=(100,), activation=tanh, lr=0.001`	0.8632

The best SVM parameters are essentially the sklearn defaults, confirming the RBF kernel at $C = 1$ was already optimal. MLP improved greatly from tuning (F1: 0.833 $\rightarrow$ 0.861), demonstrating high sensitivity to architecture choices.

7.3 Tuned Model Comparison

Table 7.3 and Figure 7.3 show tuned performance. RF achieves the best accuracy (0.875), precision (0.891), and F1 (0.887).

Table 7.3: Tuned model performance on the test set.

Model	Accuracy	Precision	Recall	F1
RF (Tuned)	**0.8750**	**0.8911**	0.8824	**0.8867**
SVM (Tuned)	0.8696	0.8824	0.8824	0.8824
MLP (Tuned)	0.8478	0.8700	0.8529	0.8614

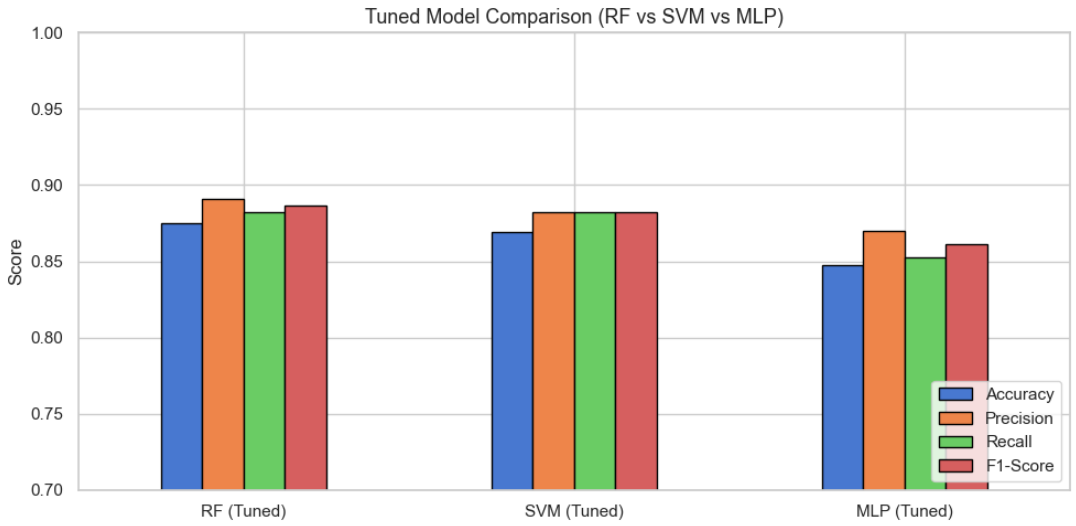


Figure 7.3: Performance comparison of all three tuned models.

7.4 Hyperparameter Effect Study

Figure 7.4 shows RF F1 improves up to $\sim$ 100 trees then plateaus, with the optimal at 200. Figure 7.5 shows SVM peaks at $C = 1$ with degradation at both extremes. Figure 7.6 shows a single layer $(100,)$ outperforms all deeper architectures, more layers overfit on this dataset size.

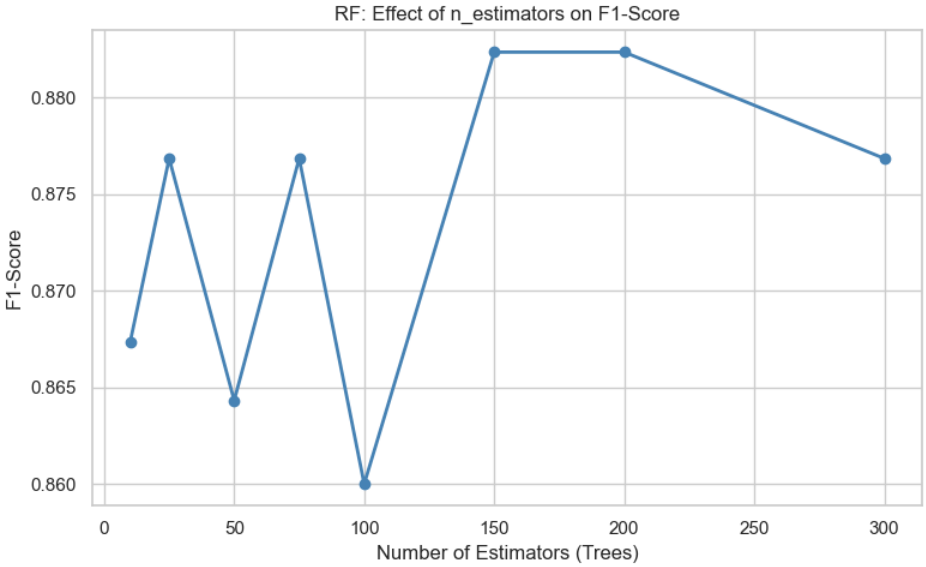


Figure 7.4: Effect of `n_estimators` on RF F1. Performance plateaus beyond 100.

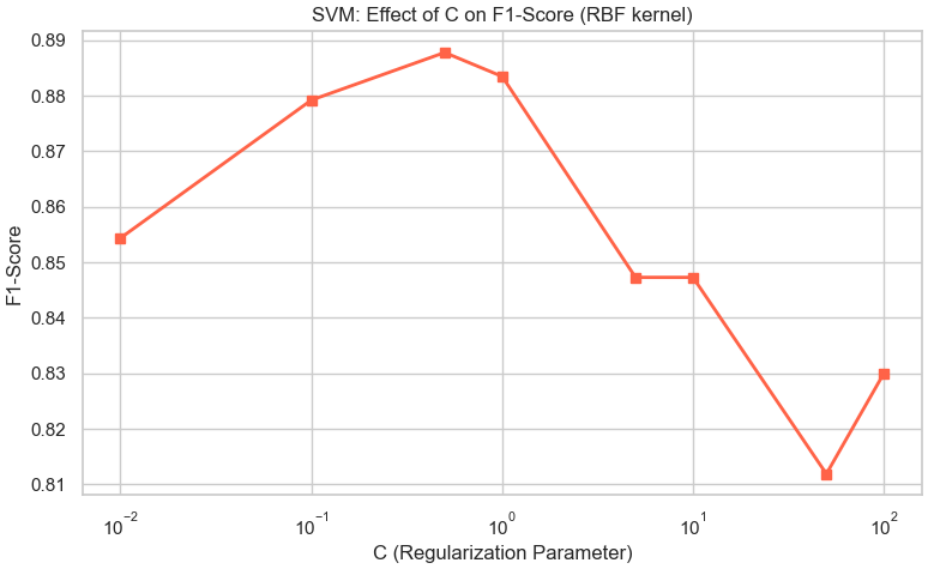


Figure 7.5: Effect of SVM regularisation parameter C (log scale). Optimal at $C = 1$.



Figure 7.6: Effect of MLP architecture. Single layer $(100,)$ outperforms deeper nets.

7.5 Cross-Validation Comparison

Table 7.4 and Figure 7.7 show 5-fold CV results. RF achieves the highest mean F1 (0.8693). MLP has the lowest variance ($\sigma = 0.0154$), indicating the most consistent generalisation across folds. SVM shows the highest variance (0.0223), suggesting more sensitivity to which samples are in each fold.

Table 7.4: 5-fold CV F1-score statistics for all tuned models.

Model	Mean F1	Std Dev
RF (Tuned)	0.8693	0.0205
SVM (Tuned)	0.8625	0.0223
MLP (Tuned)	0.8632	**0.0154**

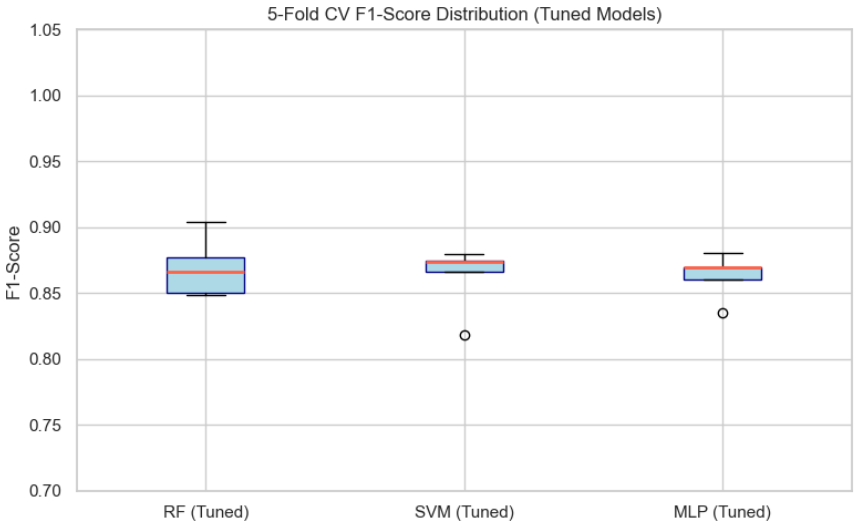


Figure 7.7: 5-fold CV F1 distributions. RF highest median; MLP tightest spread.

7.6 Computational Efficiency

Table 7.5, Figure 7.8, and Figure 7.9 report training time, prediction time, and peak memory usage measured using `tracemalloc` for each tuned model. `tracemalloc` is Python's standard-library memory profiler; it records the peak Python-heap allocation during model fitting.

Table 7.5: Full computational efficiency comparison for all tuned models.

Model	Acc.	F1	Train (s)	Predict (ms)	Peak Mem (MB)
RF (Tuned)	**0.875**	**0.887**	0.965	58.55	0.36
SVM (Tuned)	0.870	0.882	**0.039**	17.54	**0.18**
MLP (Tuned)	0.848	0.861	3.284	**3.99**	0.78

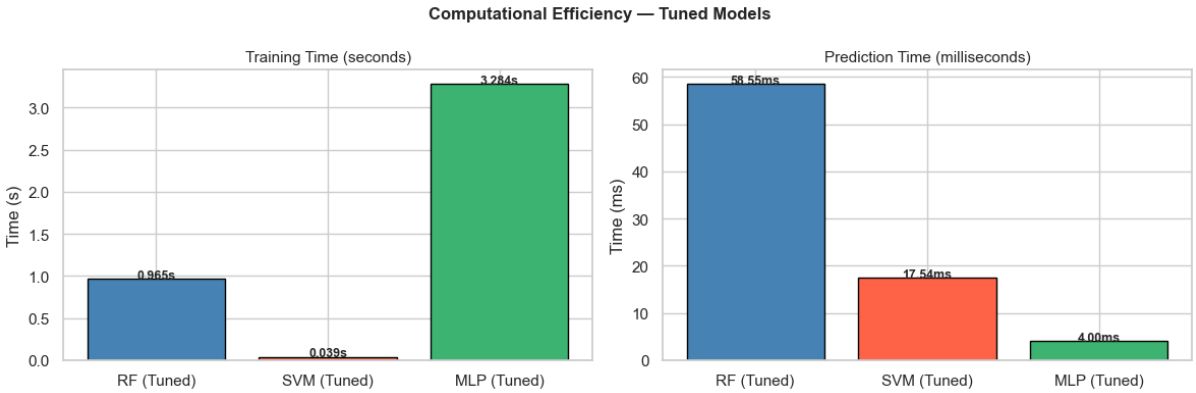


Figure 7.8: Training time in seconds (left) and prediction time in milliseconds (right) for all tuned models.

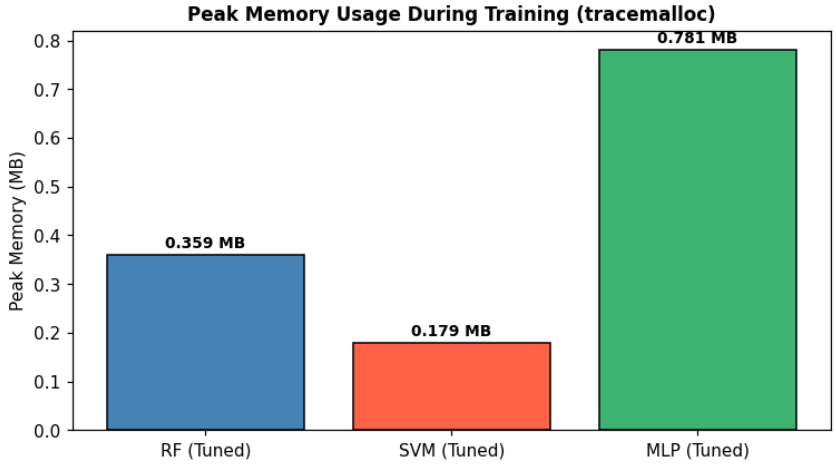


Figure 7.9: Peak memory usage during training measured with `tracemalloc`.

SVM is the most computationally efficient model: it trains 25× faster than RF and 84× faster than MLP, while consuming the lowest peak memory (0.18 **MB**). Given its F1-score of 0.882, only 0.005 below RF, SVM offers the best accuracy-to-cost trade-off and is the preferred choice when computational resources are constrained. MLP predicts fastest at 3.99 ms per batch, advantageous for real-time inference, but it has the highest peak memory (0.78 MB) due to the weight matrices and gradient buffers maintained during backpropagation. RF sits in the middle on memory (0.36 MB), note that `tracemalloc` captures only Python-level allocations, so the numpy/C-level arrays storing the ensemble's 200 trees are not fully reflected in this number.

7.7 Strengths and Limitations

Table 7.6: Strengths and limitations of each model.

Model	**Strengths**	**Limitations**
Random Forest	Best accuracy and F1 (0.875, 0.887); robust to outliers; native feature importance; no scaling required; stable CV (σ = 0.021)	Slowest prediction (58 ms); reduced interpretability; moderate Python-heap memory (0.36 MB by `tracemalloc`) with additional C-level storage for trees
SVM	Fastest training (0.039 s); lowest peak memory (0.18 MB); strong default performance; well-founded margin theory	Requires feature scaling; $O(n^2)$–$O(n^3)$ training complexity on large datasets; kernel choice adds complexity
MLP	Fastest prediction (3.99 ms); universal function approximator; improves greatly with tuning (F1: 0.833 $\rightarrow$ 0.861)	Slowest training (3.28 s); highest peak memory (0.78 MB); highest tuning effort; sensitive to initialisation; least interpretable

8 Conclusion

This study conducted a complete machine learning pipeline on a Heart Disease dataset, covering exploratory analysis, data cleaning, feature engineering, class balancing, and comparative model evaluation across three classifiers.

The EDA phase uncovered five distinct data quality issues: explicit missing values in `ChestPainType` (80 records) and `MaxHR` (49 records), noisy entry-error codes in `Sex` (10 records), physiologically impossible zeros in `Cholesterol` (172 records) and `RestingBP` (1 record), and statistical outliers in `Oldpeak` (max = 15.0). Each was addressed with a domain-appropriate technique. The preprocessing pipeline reduced the feature space from 12 raw features through 18 encoded dimensions down to 11 carefully selected features, applying three complementary selection methods: variance filtering, correlation-based redundancy removal, and embedded RF importance ranking.

Among the three classifiers, the tuned **Random Forest** achieved the best predictive performance with accuracy 0.875, recall 0.882, and F1-score 0.887. **SVM** demonstrated the best accuracy-to-cost trade-off: F1 = 0.882 at only 0.039 s training time and the lowest peak memory (0.18 MB), making it the practical choice under resource constraints. **MLP** improved most from tuning (F1: 0.833 $\rightarrow$ 0.861) but remained the slowest to train (3.28 s) and the most memory-intensive (0.78 MB) relative to its accuracy gain.

The hyperparameter studies confirmed clear optimal regions for each model's key parameters. Cross-validation confirmed stable generalisation for all tuned models, with RF leading at mean F1 = 0.869 and MLP showing the most consistent variance ($\sigma = 0.015$). The most discriminative features, `ST_Slope_Flat`, `MaxHR`, and `Oldpeak`, align with established clinical knowledge on cardiac stress response, strengthening confidence in the reliability of the trained models.

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